

LEARNING PLAN							
Unit of Competence: BASIC ELECTRONICS		Unit Code: IT/CU/ICTA/CC/01/5/MA		Trainer’s Name: Margaret waithera			
				INSTITUTION: ABERDARE TECHNICAL COLLEGE			
Date of Preparation: 24/4/2026		Date of Revision:		Class: Sept/2025			
Course: ICT		Department: ICT		Level: 5			
Term:1				No. of trainees:			
Job task/skill: This unit specifies the competencies required to apply basic electronic. It involves identifying electric circuits, identifying electronic components, applying semi-conductor theory, and classifying computer memory, applying logic gates, applying logic gates and performing Boolean algebra operations.							
Benchmark criteria: 1. Identify electric circuits Duration 2. Identify electronic components 3. Apply semi-conductor theory 4. Classify computer memory 5. Apply logic gates 6. Perform Boolean algebra operations							
	Session	Session Title	Learning Outcomes	Key points	Trainee Activities	Resources/References	Date

1	1	Identify electrical circuits	By the end of the session, the trainee should be able to - Identify Electrical circuits - Identify Components of electrical circuit - Explain Electrical quantities and their S.I units'	electrical circuit Components Emf Current Power	Trainee to: Demonstration Group discussions Observation	Floyd, T. L. (2021). <i>Electronic devices</i> (11th ed.). Pearson. Boylestad, R. L., & Nashelsky, L. (2019). <i>Electronic devices and circuit theory</i> (12th ed.). Pearson. real case examples		
2	1	Identify Electronic components	By the end of the session, the trainee should be able to identify: - Identify electronic components - Explain Characteristic of electronic components - Discuss Application of electronic components.	Resistor Capacitor Diode Inductor	Demonstration Group discussions Observation	AIDS: Resistor Capacitor Diode Inductor Layout charts, samples, whiteboard		

Week	Session	Session Title	Learning Outcomes	Key points	Trainee Activities	Resources/References	Date
3&4	`1	Apply semi-conductor theory	By the end of the session, the trainee should be able to: • Explain semiconductor theory • Describe structure of matter • Explain Electrons in conductors and semiconductors	semiconductor Silicon germanium	Demonstration Group discussions Observation	Layout charts, samples, whiteboard	
5&6	1		• Explain P-type and N-type materials • Describe of P-N junction diodes	P-type and N-type P-N junction diodes	Demonstration Group	Floyd, T. L. (2021). <i>Electronic devices</i> (11th ed.). Pearson. Boylestad, R. L., & Nashelsky, L. (2019).	

			- Describe Types and operations of transistors - Explain Application of Semiconductor theory	Forward biasing Reverse biasing PNP type NPN type	discussions Observation	<i>Electronic devices and circuit theory</i> (12th ed.). Pearson.	
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7	1	Classify computer memory	By the end of the session, the trainee should be able to: - Identify computer memories. - Classify computer memory. - Identify Memory hierarchy speed	Primary memory Secondary memory RAM ROM DAM	Practical Activities Project work Demonstration	Layout charts, samples, whiteboard	

8&9	1	Apply logic gates	- Identify Logic gates - Explain the development of Logic circuits - Simplify Logic circuits	AND Gate OR Gate NOT Gate NAND Gate NOR Gate Boolean Algebra	Demonstration Group discussions Observation		
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10&11	1	Perform Boolean algebra operations	By the end of the session, the trainee should be able to - Explain Key concepts in Boolean algebra - Demonstrate Boolean expressions - Perform Basic Boolean operations	K-Maps Software and CAD Tools Quine-McCluskey Algorithm	participate Hands on activity Demonstration Group discussions Observation	Layout charts, samples, whiteboard	

12	1		• simplify Boolean expressions • Illustrate Boolean Laws and Theorems • Simplify rules for Boolean expressions	K-Maps Software and CAD Tools Quine-McCluskey Algorithm	Demonstration Group discussions Observation	Floyd, T. L. (2021). <i>Electronic devices</i> (11th ed.). Pearson. Boylestad, R. L., & Nashelsky, L. (2019). <i>Electronic devices and circuit theory</i> (12th ed.). Pearson. Layout charts, samples, whiteboard	

Week	Session	Session Title	Learning Outcomes	Key points	Trainee Activities	Resources/References	Date
13	EXAM						

Prepared by: MARGARET WAITHERA

Date.....Sign.....

Verified by.....

Date.....Sign.....

